

Effect of a snow mass passing through an open-type cross-flow turbine on the power generation

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ABSTRACT

Small-scale hydropower generation is increasingly recognized as a distributed power source with low environmental impact and high disaster resilience. In Japan, snow accumulation in snowy regions can be carried by water flows, impacting the performance of microhydroturbines. This study performed field experiments using spherical snowballs to replicate these snow masses. This study aims to evaluate the response of an open-type cross-flow hydraulic turbine to snowball interference under winter field conditions. Snowball was released upstream of the turbine, and detailed observations were made of both the turbine's output variations and the snowball behavior. The experiments revealed that when the snowball diameter was approximately equal to the rotor blade spacing, the power output temporarily decreased but quickly recovered. Conversely, when the snowball diameter exceeded the blade spacing, the power output experienced sharp decline, with recovery times varying across experiments. It was also confirmed that the compression, crushing, and downstream discharge of the snowball by the blades influenced the output fluctuations. These findings provide initial insight into the behavior of open-type microhydraulic turbines under snow mass interference, which is relevant to winter operation in snowy regions.

1. Introduction

Small-scale hydropower generation using energy from small rivers is attracting attention [1,2] because it eliminates the need for large infrastructure such as reservoirs and conduits, thereby minimizing environmental impact. Moreover, it can function as a disaster-resistant distributed power source, supporting local electricity production and consumption without relying on the power grid. Recently, there has been a growing number of reports on using underexploited hydroelectric energy from small rivers, agricultural canals, industrial drainage systems, and water supply facilities [3,4]. To accommodate different flow rates, heads, and channel configurations, a range of small hydroturbines (known as microhydroturbines) have been developed. Examples include cross-flow [5,6], propeller [7,8], Savonius [9,10], Darrieus [11, 12], and impulse types [13,14]. These developments align with efforts

towards achieving carbon neutrality and sustainable development goals (SDGs).

Approximately 73% of the Japanese archipelago is covered by mountainous terrain with numerous steep slopes. With annual precipitation approximately 1.5 times the global average, Japan is endowed with abundant water resources. These geographical and climatic conditions generate substantial hydro energy. The authors [15] noted that regions in Japan with abundant technically and economically viable hydro energy often coincide with areas that experience heavy annual snowfall. For example, Shinano-machi in Nagano Prefecture is one of Japan's prominent heavy snowfall areas. Fig. 1 shows an agricultural canal photographed in winter, measuring 700 mm in width and depth with a 0.4-degree gradient. Figs. 1(a) and 1(b) show snow accretion and cornice forming on the canal's side walls. These formations, influenced

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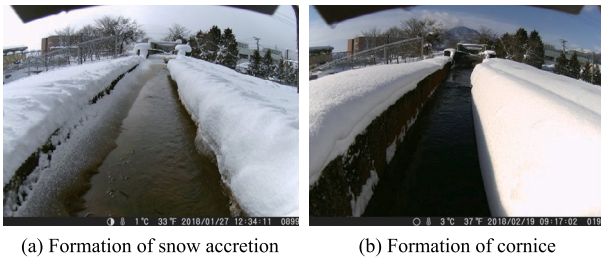


Fig. 1. Irrigation canal during the snow season in one of Japan's regions with the heaviest snowfall.

by meteorological factors such as temperature and wind speed, eventually fall into the canal, becoming snow masses that flow downstream. Such snow-laden water flows are common in snowy regions, where microhydropower turbines often encounter these snow masses, leading to reduced rotor speeds, power generation failures, and potential rotor damage. Reduced rotor speeds can hinder water flow in the canal, potentially leading to overflow. To fully harness hydro energy in snowy regions year-round, it is crucial to develop microhydropower turbines that can operate efficiently despite the presence of snow masses. Achieving this would greatly expand the use of small-scale hydro energy in snowy areas across the globe.

The authors [15,16] previously conducted field experiments during winter to develop a snow-resistant microhydropower turbine. These experiments used the agricultural canal depicted in Fig. 1, where an open-type cross-flow hydro turbine was installed. Spherical snowballs were used to simulate snow masses and were released upstream of the hydro turbine to observe changes in turbine output [15]. The results indicated that when the snowball's diameter exceeded the rotor blade spacing, the snowball became entrained in the rotor, leading to a reduction in rotor speed and output. Furthermore, it was found that when the rotor remained stationary for a certain period, an "output overshoot phenomenon" occurred upon restarting, where the output temporarily surged beyond the normal level. This revealed the relationship between the energy lost during rotor stoppage and the energy gained during the overshoot [16]. While the initial movement of the snowball upon contact with the rotor was observed, the complete behavior of the snowball, including its crushing and discharge by the blades as it passed through the rotor, remains unclear. Additionally, because the experiments involved releasing snowballs at regular intervals to simulate consecutive snow masses, a comprehensive understanding of how a single snowball interacts with the rotating blades—key to elucidating the nature of this interaction—has not yet been achieved. Furthermore, the effect of the gap between the rotor perimeter and the canal bed, a critical installation factor, on turbine output changes during snowball descent is still not fully understood.

In snowy regions, large amounts of snow often fall into agricultural canals due to collapsing snow cornices. These snow masses differ from common debris in both size and frequency. Installing protective meshes upstream is not always practical, as they can clog easily, leading to elevated upstream water levels or even flooding. Therefore, this study simulates the direct interaction between snow masses and the rotor to evaluate turbine behavior and power fluctuations under such harsh conditions. In this study, winter field experiments were performed using the same canal and open-type cross-flow hydro turbine as in previous research [15,16]. To address these issues, a single spherical snowball was released upstream of the turbine, and its behavior was observed from initial collision with the blades to compression, cutting, and downstream discharge. The relationship between these behaviors and changes in turbine output was analyzed. Key experimental parameters included the snowball diameter and the gap between the rotor perimeter and the canal bed. This approach identified the fundamental phenomena occurring as snowballs passed through the rotor, offering



Fig. 2. Photograph of the irrigation canal shown in Fig. 1 taken during the snow-free season.

insights into the reduction and cessation of turbine output owing to snowball from a more fundamental perspective. These findings provide valuable design guidelines for creating microhydropower turbines that are resistant to snow and ice.

2. Irrigation canal and open-type cross-flow hydraulic turbine

2.1. Overview of the agricultural canal

Fig. 2 shows an image of the agricultural canal used in this study during a non-snowy period. This canal was also used in previous studies [15,16]. Its winter condition is depicted in Fig. 1. The canal is built from U-shaped reinforced concrete, measuring 700 mm in both width and depth, with a gradient of 0.4 degrees. Canals of this type are common throughout Japan, with a total length of approximately 400,000 km—equivalent to circling the Earth ten times.

According to the Japan Meteorological Agency's Automated Meteorological Data Acquisition System (AMeDAS) [17], located approximately 200 m from this canal, the average annual snowfall from 1991 to 2020 was 7.26 m. Additionally, the authors independently measured the canal's average annual water level and flow rate at 80 mm and 0.06 m³/s, respectively.

2.2. Overview of open-type cross-flow hydraulic turbine

In agricultural canals in regions with heavy snowfall, water flow is typically characterized by low water levels and high velocities. These shallow, fast-moving flows are well-suited for power generation with open-type cross-flow hydro turbines [18]. In previous studies [15,16], an open-type cross-flow hydro turbine was installed in a canal to investigate changes in turbine output caused by snow masses.

In this study, the same open-type cross-flow hydro turbine was used and installed in the aforementioned canal for the experiments. The generated power of up to 30 W is suitable for powering low-consumption devices such as IoT sensors, wireless communication systems, and remote environmental monitoring units. The turbine design is depicted in Fig. 3. Although visually similar to a traditional hydro turbine, the turbine used in this study is classified as an open-type cross-flow turbine. Water enters between the blades on the outer periphery of the rotor, flows through the rotor, and exits again through the blades. The diameter, blade shape, and layout of the rotor were optimized based on the flow conditions in shallow agricultural canals [6,18]. The rotor has a diameter D of 600 mm and a width B of 300 mm. An upstream collector plate, with a total length of 380 mm, includes a straight flow path of 110 mm just before the rotor. There is an 80 mm gap between the lower end of this flow path and the rotor perimeter. The collector plate's inlet width matches the canal width W (700 mm), while its outlet width is slightly narrower at 250 mm compared to the rotor width B .

Fig. 4 shows the shape and dimensions of the rotor, which features 18 blades. The blade spacing L at the rotor perimeter is $0.175D$, and the blade angle β is 24 degrees. The blades are arc-shaped, with a chord length of 80 mm and a thickness of 2.5 mm. Both the rotor and the blades are constructed from stainless steel.

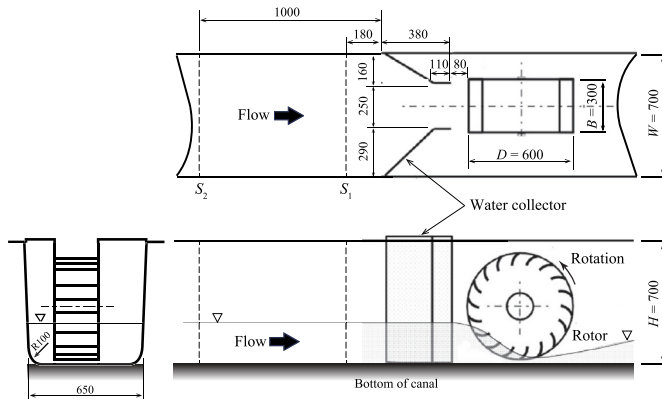


Fig. 3. Geometry and dimensions of the open-type cross-flow hydroturbine.

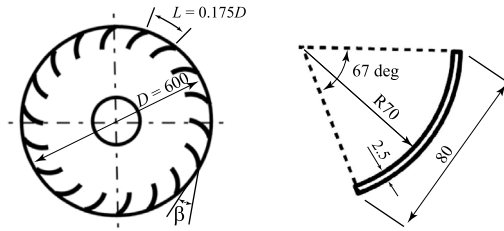


Fig. 4. Rotor and blade of the open-type cross-flow hydroturbine.

3. Experimental method and conditions

3.1. Experimental method

The turbine's rotational speed was evaluated under naturally varying conditions determined by the applied resistive load. For each case, the resistance was fixed, and the corresponding voltage and RPM were recorded simultaneously to calculate the power output.

A permanent magnet synchronous generator with 40 poles is directly coupled to the rotor's shaft. The rotor speed N is regulated by adjusting the resistance R of a load connected to the generator, with R varying from 16 to 1500 Ω . The generator's output waveform from is recorded via an A-D converter and transferred to a computer at a data sampling frequency of 100 Hz. Rotor speed N is calculated by measuring the zero-crossing period of the waveform. The dimensionless tip speed ratio λ is defined by Eq. (1):

$$\lambda = \frac{\pi DN}{60U_0}, \quad (1)$$

where U_0 is the flow velocity at cross-section S_1 (Fig. 3) in the absence of the turbine. The velocity U_0 is determined by averaging measurements taken at nine points in cross-section S_1 using an electromagnetic flowmeter.

The effective voltage V is calculated from the recorded output waveform, and the generated power P is computed using the formula $P = 3V^2/R$. The power coefficient C_p is then defined to express P in a dimensionless form as in Eq. (2):

$$C_p = \frac{P}{0.5\rho_w H_0 W U_0^3}, \quad (2)$$

where ρ_w is the density of water, and H_0 is the water level at cross-section S_1 without the turbine installed. The experiments confirmed that the upstream flow velocity remained consistent before and after installing the turbine, indicating that the blockage effect was negligible.

For the experiments, snowballs were made by packing snow uniformly into two hemispherical molds and then joining them to form a sphere. Fig. 5 shows an image of a snowballs with a diameter of



Fig. 5. Snowballs with a diameter of $d = 240$ mm.

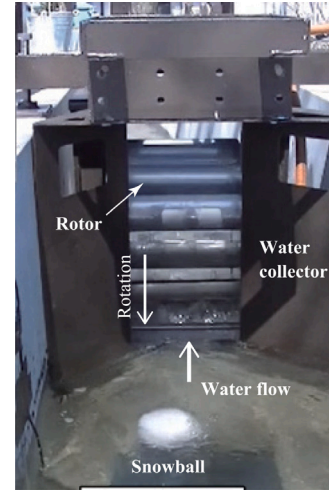


Fig. 6. Snowball moving towards the rotating rotor.

$d = 240$ mm. The hydroturbine was operated at a speed N_0 that maximizes power P_0 in clear water. A single snowball was released from the surface flow upstream of the turbine, with the release point positioned at the center of cross-section S_2 , located 1000 mm upstream of the collector (Fig. 3).

Fig. 6 shows an image of the released snowball and the turbine rotor, captured 0.5 s after release by a camera positioned 1500 mm upstream of the turbine. The snowball, seen near the water surface, is moving towards the rotor owing to buoyancy.

In a previous study [16], a camera positioned directly above the rotor was used to observe the snowballs as they entered it. Additionally, an endoscopic camera (480 × 640 pixels, 25 fps) was installed on the canal bed, 1180 mm upstream of the rotor's central axis, to capture underwater video of the snowball behavior and rotor rotation. This setup enabled a detailed analysis of the relationship with the turbine output.

Fig. 7 shows that the gap between the rotor perimeter and the canal bottom is designated as C . The rotor's central axis support structure is vertically adjustable, enabling C to be varied as needed. Experiments were performed with C set to 5, 10, and 15 mm. The blade initially struck by a descending snowball is labeled b1 (Fig. 7), with subsequent blades contacted as the rotor rotates clockwise labeled b2, b3, and so on.

3.2. Experimental conditions

Table 1 outlines the experimental conditions. The diameter of the snowball d and the gap between the rotor perimeter and the canal bottom C were varied as experimental parameters, resulting in seven different conditions, labeled Exp.1 - Exp.7. The dimensionless values d/L and C/D are also provided. Five trials were performed for each

Table 1
Experimental conditions.

	Exp.1	Exp.2	Exp.3	Exp.4	Exp.5	Exp.6	Exp.7
d [mm]	130	210	240	130	210	240	240
(d/L)	(1.24)	(2.01)	(2.29)	(1.24)	(2.01)	(2.29)	(2.29)
C [mm]		5			10		15
(C/D)		(0.0083)			(0.017)		(0.025)

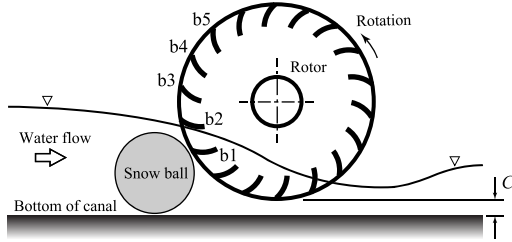


Fig. 7. Schematic diagram showing the collision of the snowball with the rotor and the definition of clearance C between the rotor and the bottom of the canal.

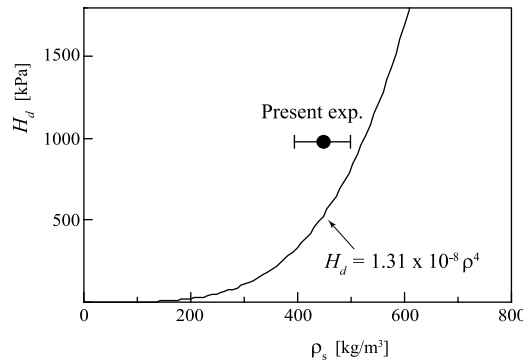


Fig. 8. Relationship between the hardness H_d and the density ρ_s of the snowball.

condition. The average density ρ_s of the snowballs across all experiments was 449 kg/m^3 , with a standard deviation of 54.8 kg/m^3 . The manufacturing error for snowball diameter was 2%, ensuring the uniformity of each snowball used. During the experiments, the ambient temperature was -1°C , the water temperature was 4°C , the flow velocity U_0 was 1.8 m/s , and the water level H_0 was 0.1 m .

Research investigating the relationship between the hardness H_d kPa and density $\rho_s \text{ kg/m}^3$ of snow from different regions [19,20] indicates that H_d is proportional to ρ_s^4 . This relationship, described by Takeuchi et al. [44], is represented by the solid line in Fig. 8 and defined mathematically by Eq. (3):

$$H_d = 1.31 \times 10^{-8} \rho_s^4 \quad (3)$$

In this study, the H_d of the fabricated snowballs was measured using a force gauge. A flat attachment with a diameter of 15 mm was inserted 15 mm into the snowball to gauge the reaction force. The average H_d measured was 0.98 MPa , as shown in Fig. 8, which is approximately 1.9 times higher than the typical value for snow. This increased hardness is attributed to the fabrication method, where snow was compacted into hemispherical molds to form the snowballs.

The snow masses flowing through agricultural canals originate from accretions and cornices accumulating over time on the canal walls and eventually falling into the canal. These snow masses are denser and harder than typical snow. Consequently, the snowballs used in this experiment effectively simulate the characteristics of these naturally occurring snow masses.

4. Results and discussion

4.1. Turbine output for clear water

The turbine's output characteristics without snowball release, i.e., in clear water, were examined. Fig. 9(a) shows the relationship between rotor speed N and turbine output P . The results for gaps C of 5, 10, and 15 mm between the rotor perimeter and the canal bed are included. Regardless of C , the maximum output P_0 is achieved at $N \approx 36 \text{ rpm}$. For $N < 36 \text{ rpm}$, P is unaffected by C . However, for speeds $36 \leq N \leq 60 \text{ rpm}$, larger values of C lead to higher P . This occurs because the collector plate, mounted on the rotor support, causes the gap between its tip and the canal bed to increase with C , thereby reducing flow resistance and increasing water velocity into the rotor. Experiments performed without a collector plate under similar flow conditions have demonstrated that increasing C reduces the contact area between the rotor and the flow, thereby decreasing the output [21]. However, in this experiment, the increase in water velocity owing to the collector plate is predominant.

Fig. 9(b) presents the dimensionless representation of rotor speed N and power output P as the tip speed ratio λ and power coefficient C_p , respectively. C_p , which indicates the efficiency of power generation, reaches a maximum value of 0.147 when $C = 15 \text{ mm}$.

4.2. Time variation of turbine output owing to a single snowball passing through the rotor

With the hydroturbine operating at speed $N_0 (= 36 \text{ rpm})$ that achieves the maximum output P_0 in clear water, a single snowball was released from cross-section S_2 upstream of the turbine (Fig. 3). The subsequent variation in turbine output P over time was then measured.

Fig. 10 shows the time variations for all experiments (Exps.1–7), as shown in Table 1. The results from five repeated experiments are presented. The change is depicted in dimensionless time $t^* (= t/T_0)$ based on the rotor period $T_0 (= 60/N_0 \text{ s})$, with the snowball release time set at $t^* = 0$. We first focus on the effect of the snowball diameter d at a clearance of $C = 5 \text{ mm}$ (i.e., Exps.1–3). For Exp.1, the output P drops sharply at $t^* = 2.92$ when the snowball contacts the blades, but it quickly recovers to the clear-water value P_0 in the repeated runs. The recovery time is approximately 1 in dimensionless terms, indicating that the snowball is discharged downstream of the rotor in one rotor revolution. The extent of the drop varies between experiments, with minima ranging from $0.21 P_0$ to $0.47 P_0$. This variation is attributable to differences in the phase angle and spanwise position of the snowball's entry onto the rotor blades. For Exp.1, the ratio d/L of the snowball diameter d to the blade spacing L is 1.24. Previous studies [16] using continuous release of snowballs with this d/L value at fixed intervals identified periodic and temporary reductions in output for each snowball.

For Exp.2 (snowball diameter $d = 210 \text{ mm}$, gap $C = 5 \text{ mm}$) in Fig. 10, P drops sharply to zero at $t^* = 2.92$ in all repeated experiments. This occurs because the snowball becomes wedged between the blades and the canal bed, causing the rotor to stop as it compresses the snowball. Following this, P remains at zero, indicating rotor stoppage, until it sharply recovers to the clear water value P_0 . This recovery happens as the snowball is crushed and reduced in size by the blades, allowing it to be discharged downstream. The time required for P to return to P_0 varies considerably between experiments owing to differences in the

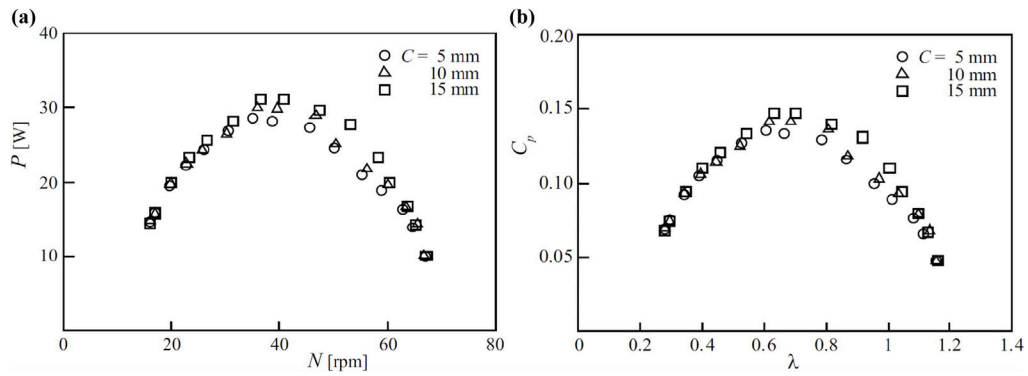


Fig. 9. Power characteristics of the rotor in clear water. (a) Power output P versus rotational speed N , (b) power coefficient C_p versus tip speed ratio λ .

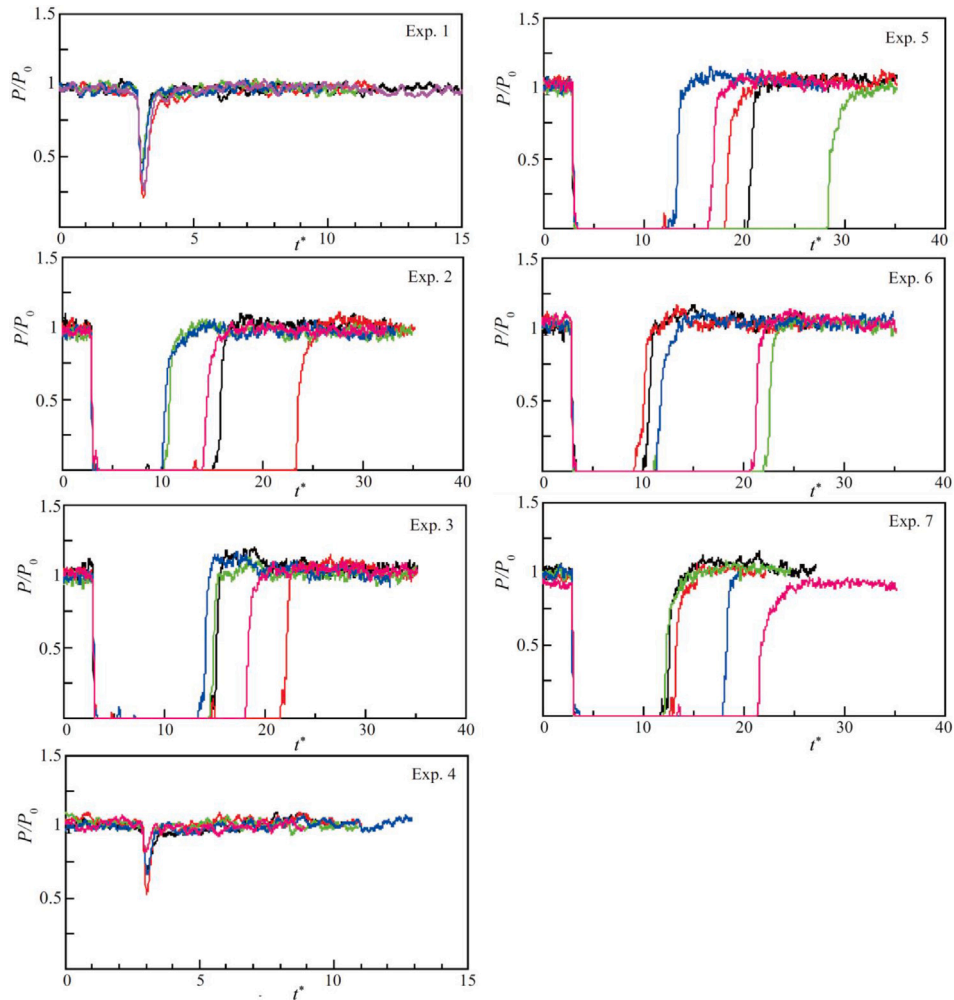


Fig. 10. Time variation of rotor power output P in Exps. 1-7.

compression process, crushing, and discharge of the snowball by the blades.

For Exp.3 (snowball diameter $d = 240$ mm, gap $C = 5$ mm) in Fig. 10, the timing of the sharp output drop is consistent across experiments. However, similar to Exp.2, the timing of the subsequent sharp rise in output varies. Occasionally, after the rise, the output temporarily exceeds P_0 before gradually stabilizing. This occurs because the rotor stoppage temporarily obstructs the upstream flow, leading to a rise in the upstream water level and a temporary surge in water flow into the rotor upon restart [16].

Exps. 4–6 show the time variation in turbine output when C is increased to 10 mm. Exp.4 shows the result of snowball diameter $d = 130$ mm. When compared to Exp.1, the output drop is similar in pattern but smaller in magnitude and shorter in duration. The increased C facilitates the discharge of snowball wedged between the blades and the canal bed more effectively. For Exps.5 and 6, which involve larger snowball diameters, the time variations exhibit behavior similar to those observed in Exps.2 and 3, respectively. The influence of C on the results is minimal.

For Exp.7 ($C = 15$ mm and $d = 240$ mm), the results show no considerable difference from those of Exps.3 and 6. Therefore, except for the case of $d = 130$ mm, which is comparable to the blade spacing L ($d/L = 1.24$), the influence of the clearance C on the output variation due to snowball passage is generally limited.

4.3. Power loss owing to a single snowball

To evaluate the turbine output reduction caused by a single snowball, the energy loss E_{loss} caused by the snowball is defined by Eq. (4):

$$E_{loss} = \int_{t_0}^{t_1} (P_0 - P) dt, \quad (4)$$

where t_0 and t_1 are the start and end times of turbine output measurement. The energy loss is normalized by the energy generated in one rotor revolution $P_0 T_0$, defining power loss ratio ϵ_{loss} calculated by Eq. (5):

$$\epsilon_{loss} = E_{loss} / P_0 T_0 \quad (5)$$

Fig. 11 shows the average power loss ratio ϵ_{loss}^* for five repeated experiments from Exps.1 to 7, plotted against snowball diameter d . For $d = 130$ mm (Exps.1 and 4), ϵ_{loss}^* is very small, below 0.24, regardless of C . This aligns with the small and temporary output drops observed in Exps.1 and 4. In contrast, for $d = 210$ mm, ϵ_{loss}^* is larger, corresponding to the extended periods of zero-output seen in Exps.2 and 5. The error bars represent the standard deviation across the five repeated experiments, highlighting considerable dispersion in ϵ_{loss}^* . For $d = 240$ mm (Exps.3, 6, and 7), both ϵ_{loss}^* and the dispersion are similarly large as for $d = 210$ mm. The influence of d and C is not considerable despite large absolute values and dispersion of ϵ_{loss}^* .

4.4. Relationship between turbine output and snowball behavior

As observed, the snowball diameter d considerably affects the time variation in turbine output P . Therefore, for $d = 130, 210, 240$ mm in Exps.1, 2, and 3, respectively, the representative time variations were chosen from five repeated experiments. An underwater camera fixed to the canal bed upstream of the rotor captured the interactions between the snowball and the blades. These selected time variations are compared in Fig. 12, with arrows indicating the capture time.

Fig. 13(a) presents underwater images from Exp. 1 ($d = 130$ mm, $d/L = 1.24$) at four different times. The blade tips and snowball are highlighted with solid and dashed lines, respectively, and the blades are identified as b1, b2, etc., as shown in Fig. 7. Supplementary diagrams show the interaction between the snowball and blades from a perspective parallel to the rotor axis. At $t^* = 2.92$, a snowball

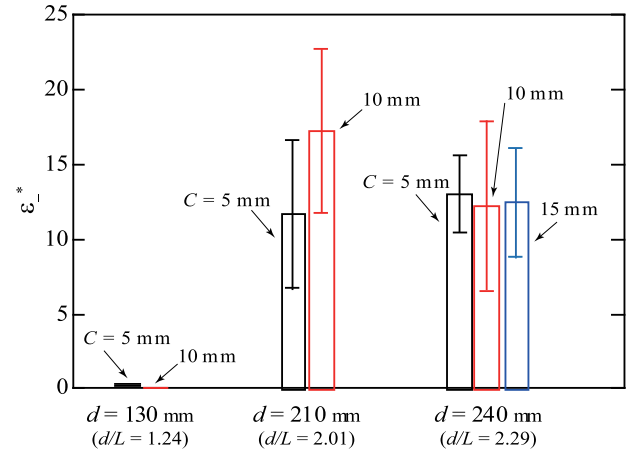


Fig. 11. Energy loss caused by the snowball.

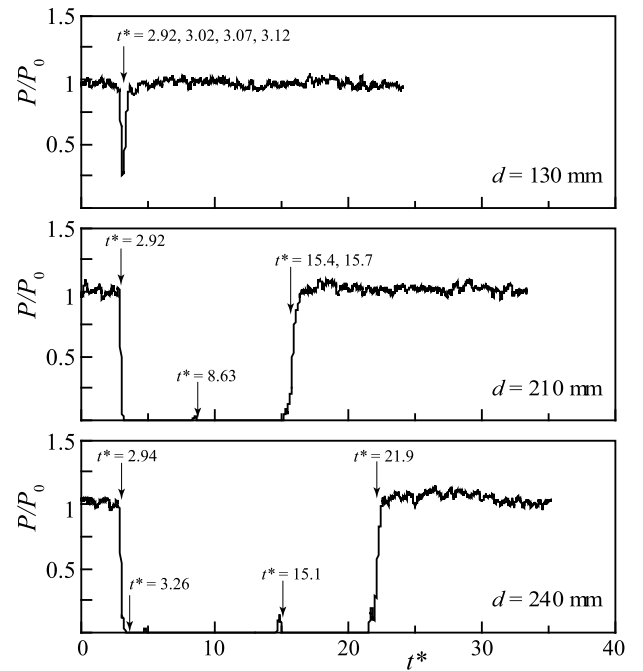


Fig. 12. Time variations of rotor power output P when $d = 130, 210$ and 240 mm.

descending towards the rotor becomes wedged between blades b1 and b2 and makes contact with the canal bed. At $t^* = 3.02$, the flow exerts counterclockwise torque on the rotor through the blades, causing blade b2 to shave the front of the snowball while blade b3 compresses it against the canal bed. This compression continues at $t^* = 3.07$. At $t^* = 3.12$, blade b3 has crushed the snowball, resulting in a minimum output of $0.28P_0$. The crushed snowball is then discharged downstream, and the output rapidly recovers. During the brief period $2.92 \leq t^* \leq 3.12$, although $P < P_0$, the rotor continues to rotate; thus, $P > 0$.

Fig. 13(b) shows images from Exp. 2 ($d = 210$ mm, $d/L = 2.01$). At $t^* = 2.92$, the snowball contacts the canal bed owing to blade b1. As blade b1 compresses the snowball against the bed, the rotor rotation stops. However, counterclockwise torque from the flow persists, causing momentary rotor movement at $t^* = 8.63$, where blade b1 shaves the snowball's front. Blade b2 then compresses the snowball, stopping the rotation once again. At $t^* = 15.4$, the snowball is eroded and reduced by the flow, with blade b3 starting to crush and discharge it downstream,

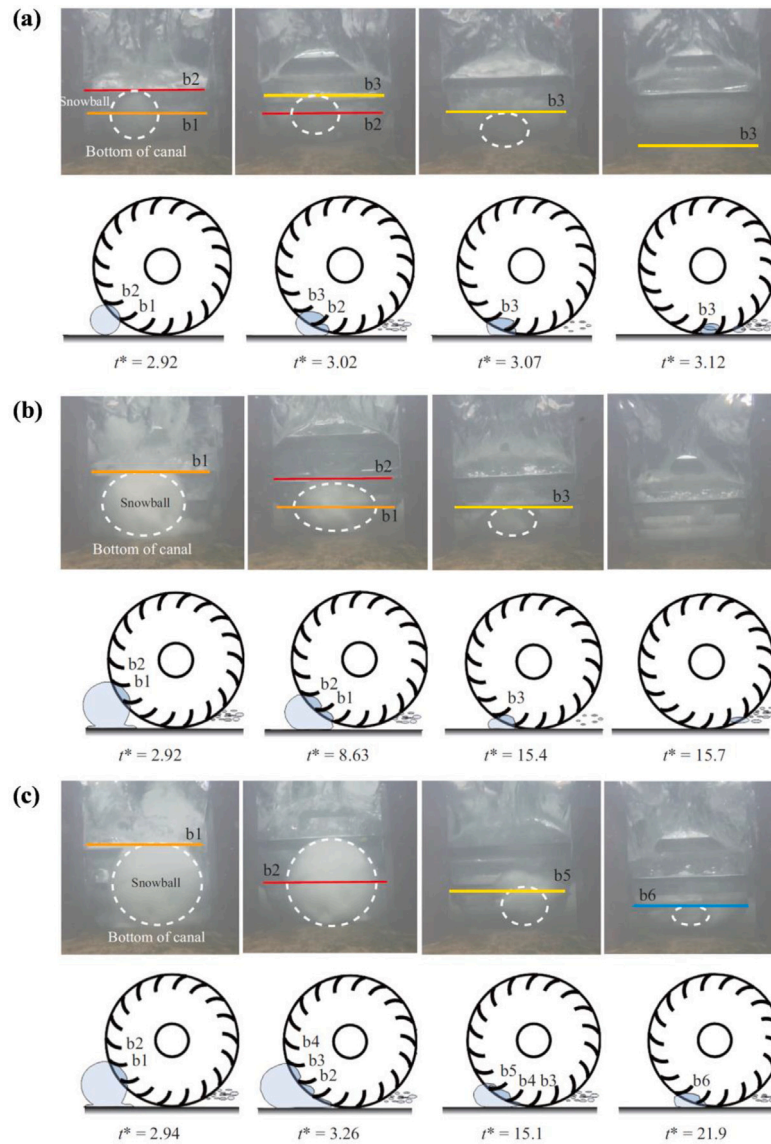


Fig. 13. Photographs of a snowball interacting with the rotor in (a) Exp. 1 ($d = 130$ mm), (b) Exp. 2 ($d = 210$ mm), and (c) Exp. 3 ($d = 240$ mm).

allowing the rotor to resume rotation. At $t^* = 15.7$, the discharge of the snowball continues, and the rotor speed or output increases.

Fig. 13(c) shows images from Exp. 3 ($d = 240$ mm, $d/L = 2.29$). At $t^* = 2.94$, the snowball contacts the canal bed, with blade b1 shaving its front. Shortly thereafter, at $t^* = 3.26$, blade b2 compresses the snowball, stopping rotor rotation. At $t^* = 15.1$, erosion has reduced the snowball's size, allowing momentary rotor movement as blades b3 and b4 shave its front. Blade b5 then compresses the snowball, stopping rotation again. At $t^* = 21.9$, blade b6 crushes and discharges the snowball downstream, leading to a rapid restoration of rotor rotation.

In previous studies conducted in the same irrigation canal [15,16], the macroscopic behavior of the hydroturbine under snow mass intrusion—such as rotor stoppage when the snowball diameter exceeds the blade spacing and the power output overshoot upon restart—was documented. However, the underlying physical mechanisms responsible for these observations remained unclear. The present study advances this understanding by releasing a single snowball and capturing its interaction with the blades using underwater imaging. This approach enabled direct observation of the fundamental sequence of events—collision, wedging against the canal bed, compression by individual

blades, shaving, crushing, and downstream discharge. These mechanisms explain the occurrence and duration of zero power output as well as the variability in recovery time seen across repeated experiments. Furthermore, this study provides the first systematic evaluation of the rotor bed clearance C . The results show that while increasing C reduces the magnitude and duration of output drops for snowballs comparable to the blade spacing, its influence is generally limited for large snowballs whose behavior is dominated by crushing and erosion processes. These findings position the present work as a mechanistic extension of previous studies and provide fundamental insights into designing hydroturbines that are robust to snow mass intrusion.

4.5. Future prospects for hydroturbine operation

This study sought to establish design guidelines for small-scale water turbines that can operate efficiently and stably in channels where snow masses flow downstream. To achieve this, field experiments were conducted in a water channel. These experiments achieved a phenomenon where the turbine performance deteriorates, leading to power output stoppage owing the interactions with the snow masses.

Detailed observations and analyses of snow mass behavior enabled a better understanding of the mechanisms underlying these occurrences.

Although the experiments were conducted using representative snowball diameters (130, 210, and 240 mm), the number of test conditions was limited. These diameters were selected based on field observations to represent typical interference scenarios involving snow mass impact on the turbine rotor. However, the data do not cover a continuous range of snowball sizes, nor do they fully represent the variability in snow-induced turbine performance degradation. Future studies should include snowballs with incrementally varied diameters to systematically characterize the turbine's power output response and to develop detailed performance curves under diverse snow mass interference conditions.

In addition, although this study used compacted snowballs based on reported densities and visual field observations, no quantitative field measurements of snow density, hardness, or size distribution were performed. Future work will involve in situ measurements of real snow and ice masses to improve the accuracy of the simulation conditions.

Regarding the clearance between the rotor and canal bed (gap C), the current study evaluated only three discrete values (5, 10, and 15 mm) owing to structural constraints. A broad and fine range of gap conditions-extending beyond the current limits should be explored in future work to clarify the impact of clearance on the flow characteristics and turbine output, especially under snow intrusion.

One potential strategy for mitigating output fluctuations is the incorporation of inertial compensation devices such as flywheels [22]. However, coupling a flywheel directly to the rotor is not a standard practice and may reduce the mechanical efficiency or increase the failure risk under sudden load changes. Instead, future implementations may consider installing a flywheel on the generator side to buffer transient fluctuations without compromising the rotor dynamics.

Furthermore, other prospective countermeasures include modifying blade shapes to minimize snow entrapment or implementing active control mechanisms to detect snow-blade contact and dynamically adjust blade orientation to release the mass downstream of the rotor. In addition, future studies will include comparative investigations of alternative rotor types (e.g., horizontal-axis Savonius and over-shot hydroturbines) to determine the most suitable configuration for shallow canal environments. Such evaluations may also help identify turbine designs that are better suited to conditions with frequent snow intrusion.

5. Conclusions

Field experiments were conducted using an agricultural canal to evaluate the performance of an open-type cross-flow hydroturbine under snowball intrusion conditions. A single spherical snowball, used to simulate a snow mass, was released upstream of the turbine, and its behavior from collision with the rotating blades to discharge downstream was observed to investigate its effects on turbine output. The snowball diameter d was selected to represent a range from sizes comparable to the blade spacing to larger sizes likely to cause significant interference. The gap C values (5–15 mm) were chosen based on structural limitations of the test apparatus.

The results indicate that when the snowball diameter d is approximately equal to the blade spacing L of the rotor, particularly when $d = 130$ mm ($d/L = 1.24$), the turbine output temporarily decreases but rapidly recovers to its clear water value. The recovery time is nearly equal to the rotor's rotation period. Conversely, when d exceeds L , specifically at $d = 210$ mm and $d = 240$ mm ($d/L = 2.01$ and 2.29), the turbine output sharply decreases, and the duration of the output stoppage varies across repeated experiments due to differences in how the blades compress, crush, and discharge the snowball.

The experimental results indicated that the influence of gap C on the power output variation was generally limited, although some sensitivity was observed depending on the snowball size and gap conditions. When

Nomenclature

B	Rotor width, m
$b1, b2, b3, b4, b5$	Blade number, –
C	Clearance between the rotor and bottom of the canal, m
C_p	Power output coefficient, –
D	Rotor diameter, m
d	Snowball diameter, m
E_{loss}	Amount of electricity lost by snowballs, J
H	Depth of the irrigation channel, m
H_d	Hardness of snow, kPa
H_0	Water level at position S_1 when the turbine is not installed, m
L	Blade spacing, m
N	Rotational speed, rpm
N_0	Rotational speed at which maximum power P_0 can be obtained when the turbine is not installed, rpm
P	Power output of the generator, W
P_0	Power output of the generator at which maximum power P_0 can be obtained when the turbine is not installed, W
R	Radius of the inside corner of the channel bottom, mm
S_1, S_2	Cross-section of the waterway, m ²
t	Time, s
t^*	Dimensionless time, –
T_0	Rotor's rotational period at N_0 , s
t_1, t_2	Turbine's output start and end times, s
W	Width of the irrigation channel, m
U_0	Flow velocity in the irrigation channel when turbine is not installed, m/s
V	Effective voltage, V
β	Inlet angle of the blade, °
λ	Tip speed ratio of the rotor, –
ρ_s	Density of snow (snowball), kg/m ³
ϵ_{loss}	Power loss ratio, –
ϵ_{loss}^*	Averaged power loss ratio, –
ρ_w	Density of water, kg/m ³

the snowball contacts the canal bed due to blade action, it is shaved and crushed by the rotating blades before being discharged downstream. For $d = 130$ mm, the output decreases to a minimum during the compression and crushing stages and recovers during discharge. However, for $d = 210$ mm and $d = 240$ mm, the snowball is compressed by multiple blades as the rotor rotates, reduced in size by the flow, crushed, and then discharged downstream. During compression, the output remains at zero and recovers during discharge.

These findings provide insights into the influence of snow mass characteristics on microhydroturbine performance and contribute to the development of snow-resistant designs.

CRedit authorship contribution statement

Kotaro Takamure: Writing – review & editing, Writing – original draft, Conceptualization. **Eiichi Satou:** Validation, Software, Formal analysis, Data curation. **Toshihiko Ikeda:** Conceptualization. **Tomomi Uchiyama:** Writing – original draft, Supervision, Conceptualization. **Tomoko Okayama:** Conceptualization. **Tomoki Miyazawa:** Data curation, Conceptualization. **Daisuke Tsunashima:** Investigation, Conceptualization. **Tomohiro Degawa:** Conceptualization. **Keigo Sasaki:** Validation, Conceptualization. **Kumiko Nagase:** Validation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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